# Implementation and Performance Evaluation of an NFT Marketplace on the Aptos Blockchain

## Chapter 1: Introduction

### 1.1 Background

Blockchain technology has significantly transformed the management of digital assets by enabling decentralized ownership, transparent transactions, and tamper-resistant record keeping. Among its most successful applications are **Non-Fungible Tokens (NFTs)**, which represent unique digital assets with verifiable ownership stored on a blockchain. NFTs have found widespread adoption in digital art, gaming, entertainment, intellectual property management, and decentralized finance.

The growing popularity of NFTs has increased the demand for decentralized marketplaces where users can mint, buy, sell, and manage digital assets without relying on centralized intermediaries. However, many existing blockchain platforms experience scalability limitations caused by sequential transaction execution, network congestion, and increasing transaction costs. These limitations directly affect the usability of NFT marketplaces, particularly during periods of high transaction volume.

Aptos is a next-generation Layer-1 blockchain designed to address these challenges through high-performance transaction processing, secure smart contract execution, and parallel transaction execution. Built using the **Move programming language**, Aptos introduces a resource-oriented execution model together with the **Block-STM** parallel execution engine, enabling improved scalability while preserving deterministic transaction execution.

In this project, an NFT marketplace was developed on the Aptos blockchain to evaluate its suitability for decentralized digital asset trading. The marketplace implements the complete NFT lifecycle, including NFT creation, listing, purchasing, and delisting, while leveraging Aptos' secure resource model to maintain ownership integrity throughout transaction execution.

### 1.2 Motivation

The motivation for selecting Aptos stems from its emphasis on scalability, security, and developer-friendly smart contract development.

Unlike conventional account-based blockchains, Aptos employs the Move programming language, which models digital assets as resources that cannot be copied, implicitly destroyed, or accidentally duplicated. This resource-oriented programming model significantly reduces the likelihood of common smart contract vulnerabilities while simplifying ownership management for digital assets.

Another distinguishing feature of Aptos is its **Block-STM** execution engine, which allows multiple independent transactions to execute in parallel while preserving deterministic results. This capability enables higher throughput without sacrificing transaction correctness, making Aptos particularly attractive for decentralized applications that process large numbers of independent transactions.

These architectural characteristics make Aptos an excellent platform for implementing secure, scalable, and efficient NFT marketplaces.

## 1.3 Objectives

The primary objective of this report is to present the implementation, deployment, and performance evaluation of an NFT marketplace developed on the Aptos blockchain.

The specific objectives include:

- Design and implement NFT marketplace smart contracts using Move.
- Develop modules supporting NFT creation, listing, purchasing, and delisting.
- Successfully deploy the marketplace on the Aptos Testnet.
- Validate all marketplace operations through functional testing.
- Evaluate performance using latency, throughput, and transaction cost measurements.
- Analyse the effectiveness of Aptos' resource-oriented architecture for NFT marketplace applications.

## 1.4 Scope of the Report

This report focuses exclusively on the Aptos implementation of the NFT marketplace. It presents the smart contract architecture, deployment procedure, functional validation, and performance benchmarking conducted on the Aptos Testnet.

Detailed security auditing and cross-platform comparisons are presented separately in the Security Audit Report and Comparative Performance Analysis Report.

# Chapter 2: Overview of the Aptos Blockchain

## 2.1 Introduction to Aptos

Aptos is a high-performance Layer-1 blockchain designed to deliver scalability, security, and low-latency transaction processing for decentralized applications. Developed using the Move programming language, Aptos builds upon resource-oriented programming principles while introducing architectural improvements that enable parallel execution of transactions without compromising deterministic behaviour.

Unlike conventional blockchain platforms that execute transactions sequentially, Aptos incorporates the **Block-STM** execution engine, allowing multiple independent transactions to execute simultaneously. The execution engine dynamically detects transaction dependencies and automatically resolves conflicts, improving throughput while maintaining consistency across validators.

The Aptos blockchain is particularly well suited for applications involving digital assets because its Move-based resource model provides strong guarantees regarding ownership, transfer, and lifecycle management of blockchain resources.

## 2.2 Resource-Oriented Architecture

The Aptos blockchain represents digital assets as **resources** stored within user accounts.

Resources possess several important properties:

- They cannot be copied.
- They cannot be discarded accidentally.
- They cannot exist without an owner.
- Ownership changes only through explicit transfer operations.

These characteristics naturally align with NFT ownership because every NFT represents a unique resource controlled by a single account.

Unlike object-oriented storage used in Sui, Aptos organizes resources within accounts while preserving strict ownership semantics through the Move language.

## 2.3 Move Programming Language

The Aptos blockchain utilizes the Move programming language to implement smart contracts.

Move provides several features that improve blockchain security:

- Resource safety.
- Strong static typing.
- Explicit ownership transfer.
- Compile-time verification.
- Deterministic execution.

Because NFTs are implemented as Move resources, ownership integrity is automatically preserved throughout marketplace operations.

Move significantly reduces the possibility of implementation errors involving asset duplication, unauthorized ownership modification, and unintended resource destruction.

## 2.4 Parallel Transaction Execution using Block-STM

One of the most significant innovations introduced by Aptos is the **Block-STM** parallel execution engine.

Instead of processing every transaction sequentially, Block-STM speculatively executes independent transactions concurrently. If conflicts are detected, only the affected transactions are re-executed while unrelated transactions continue processing.

This approach provides several advantages:

- Increased throughput.
- Lower execution latency.
- Better utilization of hardware resources.
- Improved scalability for decentralized applications.

Applications such as NFT marketplaces can benefit substantially because transactions involving different users and independent NFTs frequently execute without conflicting state updates.

## 2.5 Smart Contract Modules

The developed marketplace consists of three primary Move modules.

**NFT Module**

Responsible for NFT creation and ownership assignment.

**Marketplace Module**

Implements listing, purchasing, and delisting functionality while maintaining marketplace records.

**Reward Module**

Records user participation and demonstrates extensibility through decentralized reward tracking.

Together, these modules implement the complete NFT trading workflow on the Aptos blockchain.

## 2.6 Why Aptos for NFT Marketplaces?

Several architectural features make Aptos suitable for decentralized NFT applications.

The Move resource model naturally protects NFT ownership.

Block-STM provides high scalability through parallel transaction execution.

Compile-time resource verification improves contract reliability.

Deterministic execution guarantees consistent transaction outcomes across validators.

These characteristics enable secure management of digital assets while supporting scalable marketplace applications.

# Chapter 3: NFT Marketplace Design and Implementation

## 3.1 System Overview

The NFT marketplace developed on the Aptos blockchain provides a decentralized platform for creating, trading, and managing Non-Fungible Tokens (NFTs). The implementation leverages the **Move programming language** and Aptos' **resource-oriented architecture** to ensure secure ownership management, reliable transaction execution, and efficient handling of digital assets.

The marketplace enables users to perform the complete lifecycle of NFT transactions through the following core operations:

- NFT Minting
- NFT Listing
- NFT Purchase
- NFT Delisting

Each operation is implemented as a Move entry function and executed as an on-chain transaction. The marketplace follows a modular architecture in which individual smart contract modules are responsible for separate functionalities. This modular approach simplifies maintenance, improves code readability, and allows future extensions without affecting existing functionality.

Figure 3.1 illustrates the overall architecture of the developed NFT marketplace.

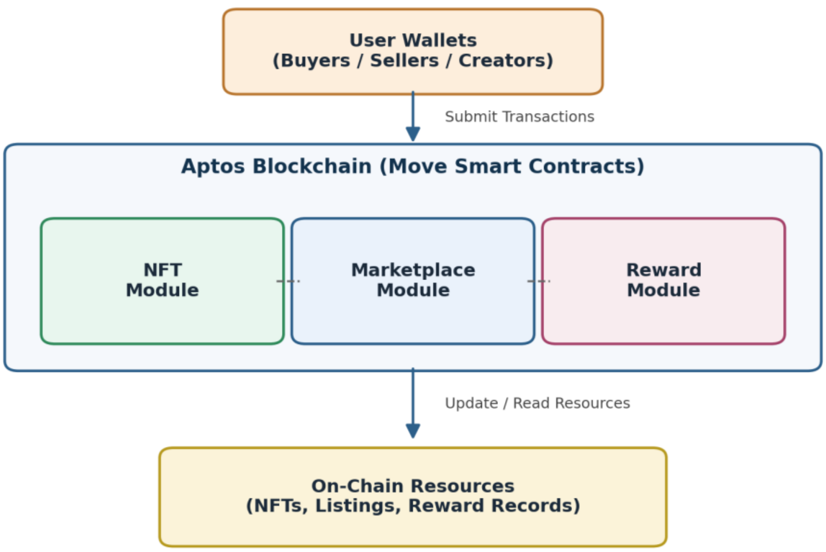



*Figure 3.1 Overall Architecture of the NFT Marketplace on Aptos*

The architecture consists of user wallets interacting with the Aptos blockchain through Move smart contracts. Marketplace operations update on-chain resources while preserving ownership and ensuring transaction consistency.

## 3.2 Smart Contract Architecture

The marketplace implementation is organized into three independent Move modules:

- NFT Module
- Marketplace Module
- Reward Module

Each module is responsible for a distinct component of the marketplace, reducing interdependencies and improving maintainability.

The interaction between these modules is shown in Figure 3.2.

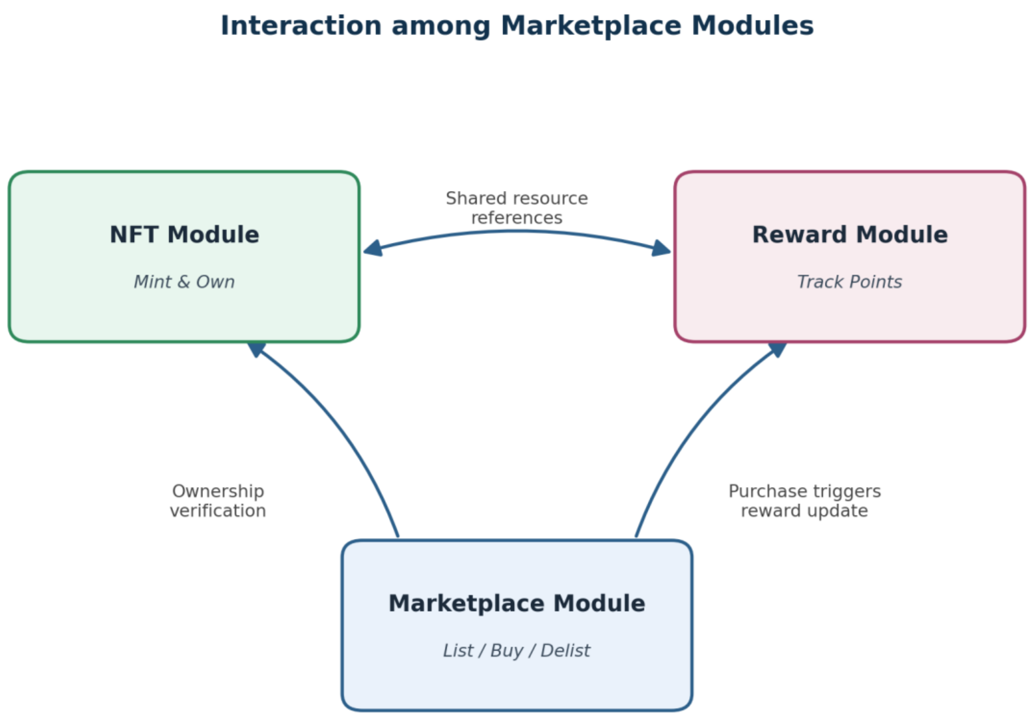



*Figure 3.2 Interaction among Marketplace Modules*

This modular design follows the principle of separation of concerns, allowing NFT management, marketplace trading, and reward management to operate independently while sharing common blockchain resources.

### 3.3 NFT Module

The NFT module is responsible for creating and managing NFT resources.

When a user mints an NFT, the module creates a new Move resource containing information describing the digital asset. The NFT includes attributes such as its unique identifier, asset name, description, creator information, and ownership details.

Unlike conventional token implementations that maintain balances within contracts, Aptos stores NFT resources directly within the owner's account. Since Move resources cannot be copied or implicitly destroyed, ownership integrity is automatically preserved by the language runtime.

The minting workflow consists of the following steps:

1. User submits a mint transaction.
2. NFT metadata is validated.
3. A new NFT resource is created.
4. Ownership is assigned to the creator's account.
5. The transaction is committed to the blockchain.

The resource-oriented nature of Move ensures that every NFT exists exactly once throughout its lifecycle.

### 3.4 Marketplace Module

The marketplace module enables users to trade NFTs securely.

Its primary responsibilities include:

- Registering NFT listings.
- Recording sale prices.
- Managing purchase transactions.
- Removing completed or cancelled listings.

When a seller lists an NFT, the marketplace verifies ownership before registering the listing. Listing information includes the NFT identifier, seller address, sale price, and listing status.

The marketplace maintains these records as on-chain resources, ensuring that every transaction operates on consistent marketplace data.

**NFT Listing Workflow**

The listing process follows these steps:

1. Seller selects an owned NFT.
2. Ownership verification is performed.
3. Listing information is created.
4. Marketplace records are updated.
5. NFT becomes available for purchase.

If ownership verification fails, the transaction is rejected without modifying marketplace state.

**NFT Purchase Workflow**

The purchase process involves coordinated execution of several operations.

First, the marketplace confirms that the requested NFT is actively listed for sale. The buyer then submits the required payment, after which ownership of the NFT resource is transferred from the seller to the buyer. Finally, the marketplace removes the completed listing from active marketplace records.

Because Aptos transactions execute atomically, either every operation completes successfully or none of the changes are committed. This guarantees marketplace consistency and prevents incomplete ownership transfers.

**NFT Delisting**

The marketplace also supports voluntary removal of NFT listings.

The delisting process consists of:

- Seller authentication.
- Ownership verification.
- Listing removal.
- Marketplace update.

Once delisted, the NFT remains under the seller's ownership and is no longer available for purchase.

## 3.5 Reward Module

To demonstrate the extensibility of the marketplace, an additional reward module was implemented.

The reward module records user participation within the marketplace and maintains reward information associated with marketplace activities. Users may accumulate reward points through successful transactions, encouraging continued engagement with the platform.

Although the current implementation provides basic reward tracking, the module establishes a foundation for future enhancements such as:

- Membership tiers.
- Trading incentives.
- Marketplace discounts.
- Reward redemption systems.

Separating reward management into an independent module improves modularity and facilitates future development.

## 3.6 Marketplace Workflow

The complete workflow of the NFT marketplace illustrates the sequence of interactions performed during a typical transaction.

The workflow proceeds as follows:

1. User creates an Aptos wallet.
2. User mints an NFT.
3. NFT resource is stored in the user's account.
4. User lists the NFT for sale.
5. Marketplace records the listing.
6. Buyer purchases the NFT.
7. Ownership is transferred to the buyer.
8. Marketplace removes the completed listing.
9. Reward information is updated.

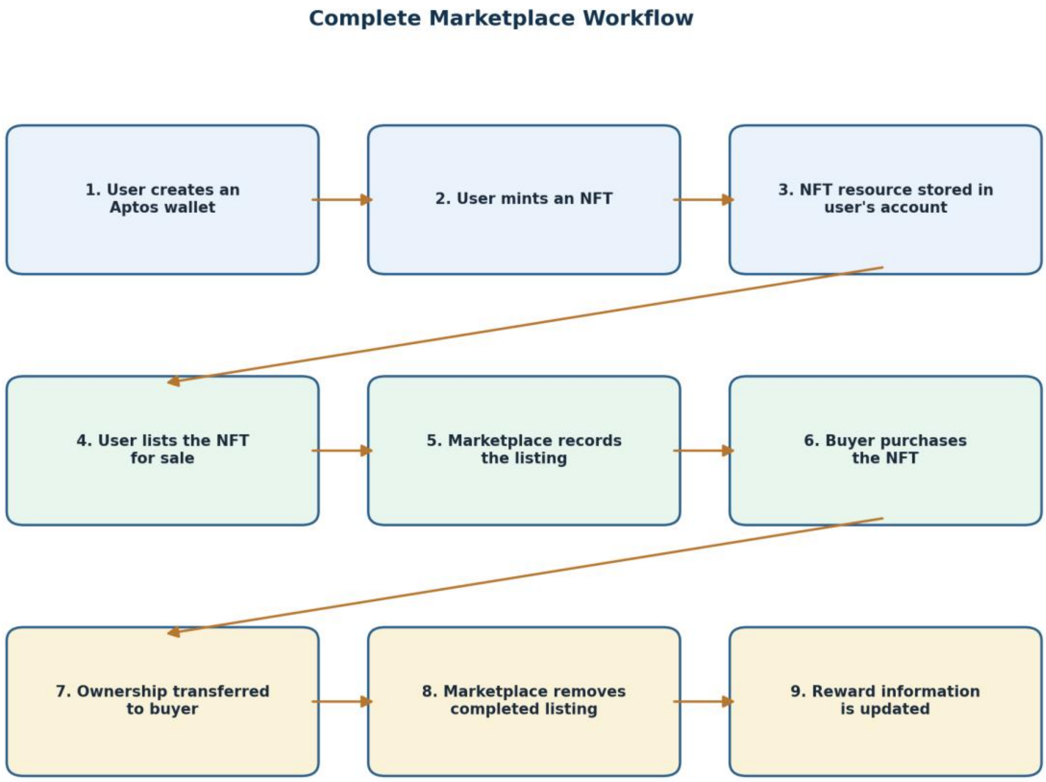



*Figure 3.3 Complete Marketplace Workflow*

This workflow demonstrates the interaction between users, smart contracts, and blockchain resources while preserving ownership throughout every transaction.

### 3.7 Design Advantages

The developed marketplace benefits from several architectural advantages provided by the Aptos blockchain.

The resource-oriented programming model naturally protects NFT ownership by preventing accidental duplication or destruction of digital assets. This significantly simplifies secure smart contract development compared to conventional account-based systems.

The modular smart contract architecture improves maintainability by separating NFT management, marketplace operations, and reward tracking into independent components.

Furthermore, Aptos' Block-STM execution engine enables parallel processing of independent transactions, providing opportunities for improved scalability under increasing transaction volumes.

Together, these design choices produce a secure, scalable, and extensible NFT marketplace suitable for decentralized digital asset trading.

# Chapter 5: Performance Benchmarking

## 5.1 Introduction

Evaluating the performance of decentralized applications is essential for understanding their practicality under real network conditions. For an NFT marketplace, performance directly influences user experience, transaction responsiveness, operational cost, and overall scalability. Metrics such as transaction latency, throughput, gas consumption, and execution reliability provide valuable insight into the efficiency of both the blockchain platform and the implemented smart contracts.

Following successful deployment and functional validation of the NFT marketplace on the Aptos Testnet, a series of benchmarking experiments were conducted. These experiments measured the execution time of the primary marketplace operations, evaluated transaction throughput under sequential execution, and recorded gas consumption for each transaction type. The objective was to establish a practical performance baseline for the developed marketplace while identifying characteristics of the Aptos execution model that influence application behaviour.

## 5.2 Experimental Methodology

Performance benchmarking was carried out using a custom benchmarking script executed through the Aptos CLI. The benchmark automated the complete marketplace workflow and executed multiple iterations of each transaction type under identical network conditions.

The benchmark evaluated the following operations:

- **Mint** – Creates a new NFT and stores it within the owner's account.
- **List** – Registers an NFT for sale on the marketplace.
- **Buy** – Transfers ownership of a listed NFT from the seller to the buyer.
- **Delist** – Removes an NFT listing from the marketplace.

Each operation was executed five times, and the minimum, average, and maximum execution times were recorded. The benchmark also executed twenty sequential transactions to evaluate practical throughput. Gas consumption values were extracted directly from transaction receipts generated by the Aptos Testnet.

Because the experiments were conducted on the public testnet using sequential CLI execution, the recorded performance includes blockchain processing time, network latency, validator confirmation, and command-line execution overhead.

## 5.3 Latency Analysis

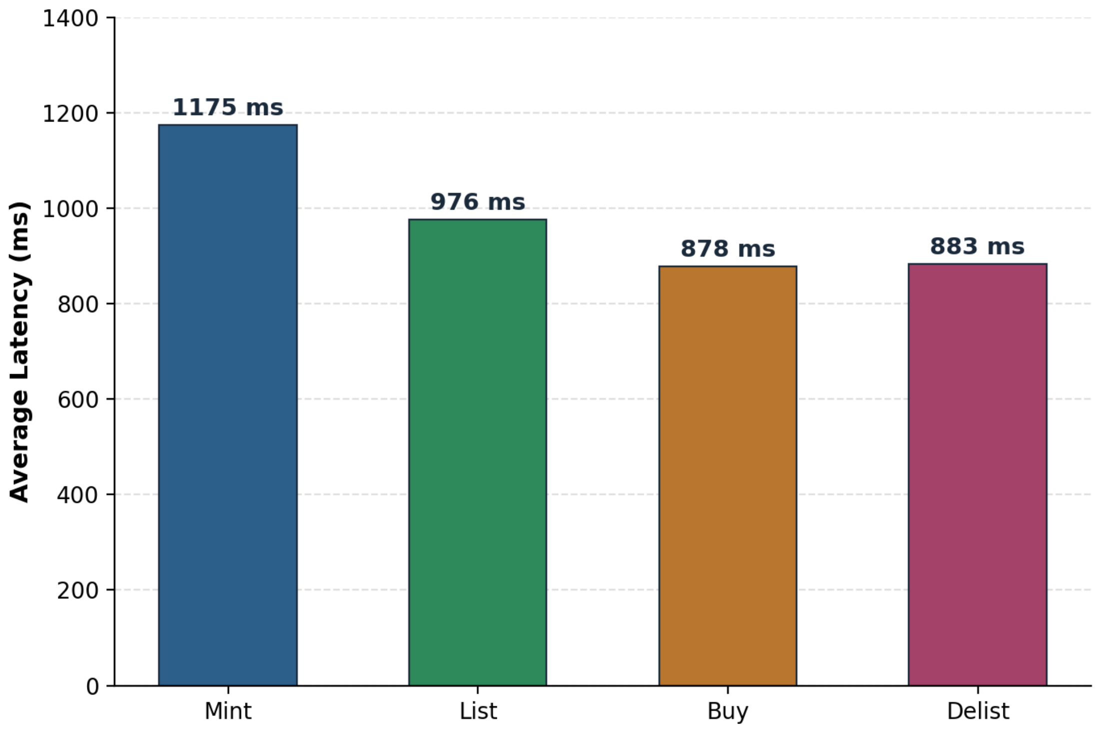
Transaction latency measures the time required for a submitted transaction to be processed and confirmed by the blockchain network. Lower latency results in faster user interactions and improved marketplace responsiveness.

The measured latency values are summarized in Table 5.1.

**Table 5.1 Transaction Latency**

| Operation | Minimum (ms) | Average (ms) | Maximum (ms) |
|---|---|---|---|
| Mint | 765 | **1175** | 2339 |
| List | 850 | **976** | 1255 |
| Buy | 758 | **878** | 943 |
| Delist | 762 | **883** | 1127 |

**Figure 5.1 Average Latency Comparison Across Marketplace Operations**




**Discussion**

The benchmark demonstrates that all marketplace operations completed within approximately one second on average, indicating responsive execution under public testnet conditions.

Among the evaluated operations, **Buy** exhibited the lowest average latency of **878 ms**. This operation primarily involves transferring ownership of an existing NFT resource and updating marketplace records, making it relatively efficient compared to operations requiring resource creation.

The **Delist** operation required an average of **883 ms**, only marginally higher than the purchase operation. Delisting simply removes marketplace information while leaving NFT ownership unchanged, resulting in minimal computational complexity.

The **List** operation completed in **976 ms** on average. This operation requires ownership verification, creation of marketplace records, and storage updates, which introduce slightly greater execution overhead.

The **Mint** operation recorded the highest average latency at **1175 ms**. During minting, a new NFT resource is created, initialized with metadata, and stored within the user's account. Resource creation and on-chain storage allocation naturally increase execution time compared with operations involving existing resources.

Overall, the observed latency values indicate that the developed NFT marketplace provides responsive transaction processing suitable for practical decentralized applications.

## 5.4 Throughput Evaluation

Transaction throughput measures the number of transactions successfully processed within a given time interval. It provides an indication of the practical transaction handling capability of the implemented application.

For this experiment, the benchmark executed twenty sequential marketplace transactions.

The observed results are presented in Table 5.2.

**Table 5.2 Throughput Benchmark**

| **Metric** | **Value** |
|---|---|
| Total Transactions | 20 |
| Execution Time | 52.3 s |
| Throughput | **0.38 TPS** |

The throughput was calculated using:

[
TPS=\frac{20}{52.3}\approx0.38
]

**Discussion**

The measured throughput of **0.38 TPS** represents the performance of sequential transaction execution through the Aptos CLI on the public Aptos Testnet. This value should not be interpreted as the maximum throughput capability of the Aptos blockchain.

Several factors contributed to the observed throughput:

- Sequential execution of benchmark transactions.
- Public testnet network latency.
- Validator confirmation delays.
- CLI execution overhead.
- Network communication between client and validators.

Aptos is designed to achieve significantly higher throughput through its **Block-STM parallel execution engine**, which allows independent transactions to execute concurrently. Since the benchmark intentionally submitted transactions one after another, the blockchain's parallel execution capabilities were not fully utilized.

Consequently, the measured TPS serves as a reproducible baseline for evaluating the implemented marketplace rather than representing the theoretical performance limits of the Aptos network.

## 5.5 Gas Consumption Analysis

Transaction cost is another important performance metric because it directly influences the operational expense of decentralized applications. Gas consumption represents the computational resources required to execute a transaction on the blockchain.

Table 5.3 summarizes the observed gas usage for each marketplace operation.

**Table 5.3 Gas Usage**

| Operation | Gas Used |
|---|---|
| Mint | 4616 |
| List | 4415 |
| Buy | 4744 |
| Delist | 49 |

**Discussion**

The benchmark results indicate that all marketplace operations consume relatively modest amounts of gas. The **Buy** operation required the highest gas usage (**4744 units**) because it combines payment processing, ownership transfer, and marketplace state updates within a single transaction.

The **Mint** operation consumed **4616 gas units**, reflecting the cost of creating a new NFT resource and storing associated metadata on-chain. The **List** operation required **4415 gas units**, demonstrating that registering marketplace information is computationally efficient.

The **Delist** operation exhibited an exceptionally low gas consumption of only **49 gas units**, indicating that removing marketplace entries involves minimal computation and storage modification. This suggests that delisting operations impose negligible operational cost compared with other marketplace activities.

Overall, the gas analysis demonstrates that the developed marketplace performs efficiently and maintains relatively low transaction costs across all core operations.

## 5.6 Performance Discussion

The benchmark results demonstrate that the implemented NFT marketplace operates efficiently on the Aptos Testnet. All evaluated transactions achieved average execution times of approximately one second or less, while every benchmark transaction completed successfully, resulting in a **100% success rate**. This confirms the reliability of the marketplace implementation under real blockchain conditions.

The observed latency patterns reflect the nature of the implemented operations. Resource creation during minting introduces additional processing overhead, whereas buying and delisting involve modifications to existing resources and therefore complete more quickly. Gas consumption remained moderate across all operations, indicating that the marketplace logic is computationally efficient and suitable for practical deployment.

Although the measured throughput is limited by sequential CLI execution, the architecture of Aptos is specifically designed to exploit **parallel transaction execution** through the Block-STM engine. Independent marketplace transactions involving different users and NFTs can therefore be processed concurrently in production environments, offering substantially greater scalability than indicated by the baseline benchmark.

# Chapter 6: Challenges, Optimization Strategies, and Conclusion

## 6.1 Challenges Encountered During Development

The implementation and deployment of the NFT marketplace on the Aptos blockchain provided valuable experience in developing decentralized applications using the Move programming language. Although the final marketplace successfully achieved all functional objectives, several technical challenges were encountered during development, deployment, and benchmarking.

One of the primary challenges involved understanding the **resource-oriented programming model** of Aptos. Unlike conventional object-oriented or account-based smart contract platforms, Move enforces strict ownership rules for resources. Every NFT is treated as a unique resource that cannot be copied or discarded, requiring careful implementation of ownership transfers during marketplace operations. Proper management of resources was essential to ensure that NFTs maintained a single owner throughout their lifecycle.

Another challenge involved implementing marketplace transactions while preserving consistency between NFT ownership and marketplace listings. Operations such as listing, purchasing, and delisting required coordinated updates to multiple on-chain resources. These updates had to be performed atomically to prevent inconsistent marketplace states or partial ownership transfers. The Move language's transaction model simplified this process by ensuring that all state changes either completed successfully or were rolled back entirely in the event of failure.

Deployment on the Aptos Testnet introduced practical considerations related to account initialization, module publication, gas budgeting, and wallet configuration. Every deployment generated a new module address that had to be referenced correctly during subsequent marketplace interactions. Proper account funding and configuration were necessary to execute transactions successfully throughout the testing process.

Performance benchmarking also presented certain limitations. The experiments were conducted through sequential command-line execution on the public Aptos Testnet. Consequently, measured latency and throughput included network delays, validator confirmation times, and command-line execution overhead. These factors produced conservative performance measurements that represent practical baseline values rather than the maximum capabilities of the Aptos blockchain.

## 6.2 Optimization Strategies

Although the implemented NFT marketplace satisfies all project requirements, several optimization strategies can further enhance its performance, scalability, and usability.

One significant optimization involves leveraging the **Block-STM parallel execution engine** more effectively. The current benchmarking methodology executed transactions sequentially, preventing Aptos from exploiting its ability to process independent transactions concurrently. Future benchmarking could employ multiple clients or concurrent transaction submission to better demonstrate the blockchain's parallel execution capabilities.

Smart contract optimization also offers opportunities for performance improvement. Reducing unnecessary storage updates, minimizing resource accesses, and optimizing transaction logic can lower gas consumption while improving execution speed. Efficient resource organization within user accounts may further reduce computational overhead during marketplace operations.

The marketplace itself can be enhanced by introducing advanced trading mechanisms such as auction-based sales, royalty distribution for creators, marketplace analytics, and reputation systems for buyers and sellers. These features would increase the practical functionality of the application while demonstrating the flexibility of Move-based smart contract development.

From an infrastructure perspective, NFT metadata and associated media files could be stored using decentralized storage solutions such as the **InterPlanetary File System (IPFS)**. This approach reduces on-chain storage requirements while preserving secure references to digital assets, improving scalability and reducing operational costs.

Finally, future performance evaluations should include stress testing under concurrent workloads, larger transaction batches, and extended execution periods. Such experiments would provide a more comprehensive understanding of the marketplace's behaviour under production-scale conditions.

## 6.3 Overall Evaluation

The implementation and evaluation presented in this report demonstrate that the Aptos blockchain provides a secure and efficient environment for decentralized NFT marketplace applications. The resource-oriented programming model of Move ensures that digital assets maintain strict ownership guarantees throughout every marketplace operation, while Block-STM offers a strong architectural foundation for scalable transaction processing.

The marketplace was successfully deployed on the Aptos Testnet, where all implemented smart contract modules operated correctly. Functional validation confirmed the successful execution of NFT minting, listing, purchasing, and delisting operations. Every benchmark transaction completed successfully, resulting in a **100% success rate**, which demonstrates the stability and correctness of the implemented smart contracts.

Performance benchmarking further indicated that marketplace operations exhibit consistently low latency and moderate gas consumption under public testnet conditions. Although the observed throughput reflects sequential execution rather than the blockchain's full parallel processing capabilities, the benchmark establishes a realistic baseline for evaluating practical application performance.

Overall, the developed marketplace effectively demonstrates how Aptos combines secure asset management with scalable transaction execution to support decentralized NFT trading applications.

## 6.4 Conclusion

This report presented the design, implementation, deployment, and performance evaluation of a decentralized NFT marketplace developed on the Aptos blockchain. The marketplace was implemented using the Move programming language and organized into modular smart contract components responsible for NFT management, marketplace operations, and reward tracking.

The smart contract package was successfully deployed on the Aptos Testnet and validated through real blockchain transactions. Functional testing confirmed the correct execution of NFT minting, listing, purchasing, and delisting while maintaining ownership integrity and consistent marketplace state throughout the transaction lifecycle.

Performance benchmarking evaluated transaction latency, throughput, gas consumption, and execution reliability. The results demonstrated responsive transaction processing, low gas consumption, and a **100% success rate** across all benchmark operations. Although the measured throughput was constrained by sequential CLI execution on the public testnet, the architecture of Aptos provides significant opportunities for higher scalability through its Block-STM parallel execution engine.

In conclusion, the developed NFT marketplace successfully fulfills the objectives of the project by providing a secure, modular, and scalable decentralized trading platform. The findings indicate that Aptos is well suited for NFT marketplace development, offering strong security guarantees through Move's resource-oriented programming model together with high-performance transaction processing capabilities that support future large-scale decentralized applications.